UG (NEP) 3rd SEMESTER
COMPUTER APPLICATIONS (MAJOR)

Session: February–March 2024 Paper: Data Communication and Networking
Time Allowed: 2:00 Hours Maximum Marks: 100 Minimum Marks: 36

NOTE: Attempt all questions from Sections A and B and only two questions from Section C.

SECTION A: Short Answer Type Questions (8 × 2 = 16 Marks)

Answer all questions in a maximum of 20 words each.

1. What is bandwidth in networking?
2. Define Nyquist sampling rate.
3. What is de-multiplexing?
4. What is modulation?
5. What is a router?
6. Write down the components of a network.
7. Define HTTP.
8. Write down the full form of UDP.

SECTION B: Medium Answer Type Questions (4 × 10 = 40 Marks)

Answer all four questions with internal choice in a maximum of 150 words each.

2. Differentiate between analog and digital signals.

OR

Explain, with examples, serial and parallel communication.

3. Explain, with the help of a diagram, the different components of Pulse Code Modulation (PCM).

OR

What are the different techniques for digital-to-analog conversion in data communication?

4. Explain different network topologies in data communication.

OR

Explain different types of computer networks.

5. Explain classful IP addressing along with its five subclasses.

OR

Explain checksum with the help of an example.

SECTION C: Long Answer Type Questions (2 × 22 = 44 Marks)

Answer only two questions in a maximum of 400 words each.

6. Explain different types of transmission media.
7. What is multiplexing? Explain different types of multiplexing in data communication.
8. Define the OSI reference model and explain its seven layers.
9. Explain, with the help of a diagram, Cyclic Redundancy Check (CRC).